COMP 3920 Database Systems 1

MySQL Full Logical Consistent Backup and Restore

Introduction:

The goal of this document is to serve as a walkthrough working with Backing up and Restoring in MySQL 8 using a full logical backup.

We frequently need to automate backups of the database in MySQL. Although we try to prevent exposing passwords in our scripts, scheduled tasks and events, sometimes the mysql password is exposed. If this password is intercepted and the mysql account is compromised, we want to limit how much access the attacker has with this account. We certainly don't want them to have access to the `root` account. Let's create a `backup` user with only the required privileges to backup the database.

At a minimum we need:

```
CREATE USER 'backup'@'localhost' IDENTIFIED BY 'backups';
GRANT SELECT, SHOW VIEW, RELOAD, REPLICATION CLIENT, EVENT, PROCESS, TRIGGER
ON *.* TO 'backup'@'localhost';
GRANT LOCK TABLES ON *.* TO 'backup'@'localhost';
```

https://www.fromdual.com/privileges-of-mysql-backup-user-for-mysqldump

We are going to use `mysqldump` to make a full logical consistent backup.

There are 3 methods to call `mysqldump`:

```
shell> mysqldump [options] db_name [tbl_name ...]
shell> mysqldump [options] --databases db_name ...
shell> mysqldump [options] --all-databases
```

https://dev.mysql.com/doc/refman/8.0/en/mysqldump.html

The first method, `mysqldump db_name [tbl_name …]` will only backup a single database (or the specified tables within a single database.

The second method, `mysqldump --databases db_name …` will backup all of the specified databases.

The third method, `mysqldump --all-databases` will backup all of the databases (excluding the system databases such as information_schema, performance_schema, mysql and sys).

We will also be backing up our mysql configuration and bin log files in case the one of these files becomes corrupt.

And lastly we need to take a copy of the users, roles and privileges. Using the tool `mysqlpump` (that's ***p***ump not ***d***ump) we will get a logical copy of the users, roles and privileges.

Step 1:

If you need a copy of the `employees`, `party` or `classicmodels` database, download `lab_databases_and_users.sql` from the Learning Hub and import the data.

Step 2:
Login to MySQL as user `root`.

Create a Backup user that has all the privileges required to perform a backup:
Here is a list of the required privileges for a backup user:

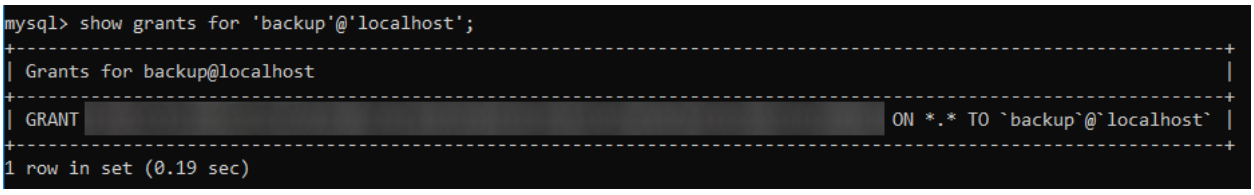
- SELECT ON *.*
- SHOW VIEW ON *.*
- RELOAD
- REPLICATION CLIENT
- EVENT
- PROCESS
- TRIGGER
- LOCK TABLES ON *.*

```
CREATE USER 'backup'@'localhost' IDENTIFIED BY 'backups';
GRANT SELECT, SHOW VIEW, RELOAD, REPLICATION CLIENT, EVENT, PROCESS, TRIGGER
ON *.* TO 'backup'@'localhost';
GRANT LOCK TABLES ON *.* TO 'backup'@'localhost';
```

Submit your screenshot of the show grant command:
`show grants for 'backup'@'localhost';`

10_Backup_user_privileges.jpg

Submit



Step 3:
Logout of MySQL.

Using `mysqldump` Backup the following databases:

1. classicmodels
2. employees
3. party
4. mysql (system table)

DO NOT backup the following databases:

- information_schema (not required)
- performance_schema (not required)
- sakila (not required)
- world (not required)
- lotsofdata (not required)

<u>Your Task:</u>
Use `mysqldump` and the ***new user*** `backup` to create a consistent, logical backup your databases. Ensure your backup command will backup **all events and routines**. Our databases are mostly InnoDB; so use the `single-transaction` switch to ensure the backup is consistent.

Here are the important command line switches:

- `--routines` (Dump stored routines (procedures and functions) from dumped databases)
- `--events` (Dump events from dumped databases)
- `--single-transaction` (InnoDB) or `--lock-tables` (MyISAM)

Redirect the output of `mysqldump` into a file with the following name:
`mysql_backup_<today's date>.sql` where <`today's date`> is in the format `YYYY-MM-DD`.
ex: `mysql_backup_2018-11-16.sql`

Submit your screenshot of the mysqldump command:
11_mysqldump_commandline.jpg

Submit

```
C:\Users\User>mysqldump -u backup -p
Enter password: *******
```

Step 4:
Create a new backup folder:
C:\Users\User\Documents\Backups\ (Windows) or
~/Documents/Backups/ (Linux)

Step 5:
Move the mysqldump backup file into your new backup folder.

Step 6:
Copy the configuration file C:\ProgramData\MySQL\MySQL Server 8.0\my.ini (Windows) to your new backup folder.

Step 7:
Backup the users, roles and privileges in the database using `mysqlpump` (Note: that's `mysql`***p***`ump` not `mysql`***d***`ump`).

Run the following command:

```
mysqlpump --exclude-databases=% --users -u backup -p >
users_backup_<today's date>.sql
```

Substitute `<today's date>` to be today's date in the format `YYYY-MM-DD`.
ex: `users_backup_2018-11-16.sql`

```
C:\Users\User>mysqlpump --exclude-databases=%  --users -u backup -p > users_backup_2018-11-16.sql
Enter password: *******
mysqlpump: [WARNING] (3719) 'utf8' is currently an alias for the character set UTF8MB3, which will be replaced
 by UTF8MB4 in a future release. Please consider using UTF8MB4 in order to be unambiguous.
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mysqlpump: [WARNING] (3719) 'utf8' is currently an alias for the character set UTF8MB3, which will be replaced
 by UTF8MB4 in a future release. Please consider using UTF8MB4 in order to be unambiguous.
Dump completed in 2413

C:\Users\User>move users_backup_2018-11-16.sql Documents\Backups
        1 file(s) moved.

C:\Users\User>_
```
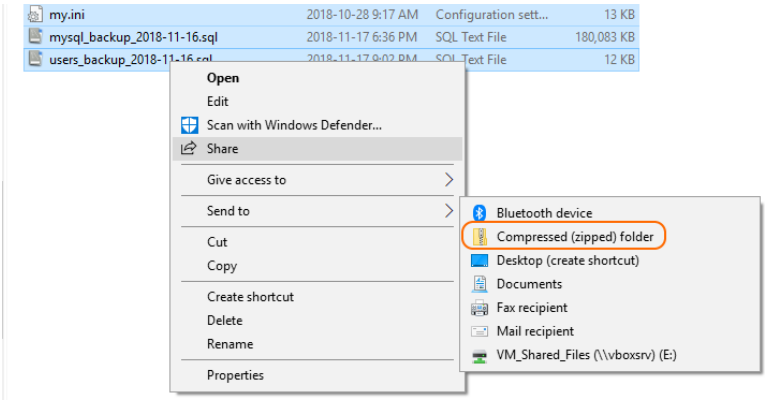

Note: You can safely ignore the warning message about UTF8, UTF8MB3 and UTF8MB4 charset aliases.

Copy the new user backup file to into your backup folder.

Step 8:
For this lab, the bin logs and data directory files will too large to submit via D2L on the Learning Hub.

Create a zip file of your Logical backup, Users backup and mysql configuration file and submit to D2L.
Do NOT include the bin logs and data directory in your zip file.



Name your backup:
`logical_backup_<today's date>.zip`

Substitute `<today's date>` to be today's date in the format `YYYY-MM-DD`.
ex: `logical_backup_2018-11-16.zip`

<u>How to Restore:</u>
You **don't need** to do these last steps. I'm providing them here in case you need them at a later point.

You can restore the databases by running mysql and redirecting the `mysql_backup` file as input.

```
mysql -u root -p < mysql_backup_<today's date>.sql
```

You can restore the users, roles and permissions by running mysql and redirecting the `users_backup` file as input.

```
mysql -u root -p -f < users_backup_<today's date>.sql
```

Notice the `-f` option for `force`. This will continue to run the script even if a single line generates an error.

If you the above command, you will likely see a few errors. Unfortunately, mysqlpump puts some of the commands to re-create the users and roles in the wrong order. In order to overcome this, we will run the script twice (with the force option) so that all the users and roles will get created.

```
user@ubuntu-VirtualBox:/media/sf_VM_Shared_Files$ mysql -u root -p -f < users_backup_2.sql
Enter password:
ERROR 3162 (HY000) at line 16: Authorization ID `hr`@`%` does not exist.
ERROR 1410 (42000) at line 17: You are not allowed to create a user with GRANT
ERROR 1410 (42000) at line 18: You are not allowed to create a user with GRANT
ERROR 3523 (HY000) at line 20: Unknown authorization ID `helen`@`%`
ERROR 3162 (HY000) at line 21: Authorization ID `hr`@`%` does not exist.
ERROR 1410 (42000) at line 22: You are not allowed to create a user with GRANT
ERROR 3523 (HY000) at line 23: Unknown authorization ID `henry`@`%`
ERROR 3162 (HY000) at line 31: Authorization ID `inventory`@`%` does not exist.
ERROR 1410 (42000) at line 32: You are not allowed to create a user with GRANT
ERROR 3523 (HY000) at line 34: Unknown authorization ID `ian`@`%`
ERROR 1396 (HY000) at line 89: Operation CREATE USER failed for 'mysql.infoschema'@'localhost'
ERROR 1396 (HY000) at line 91: Operation CREATE USER failed for 'mysql.session'@'localhost'
ERROR 1396 (HY000) at line 95: Operation CREATE USER failed for 'mysql.sys'@'localhost'
ERROR 1396 (HY000) at line 99: Operation CREATE USER failed for 'root'@'localhost'
```

Run the command **one more time** and this will complete the restore of the users and roles.

```
mysql -u root -p -f < users_backup_<today's date>.sql
```

```
user@ubuntu-VirtualBox:/media/sf_VM_Shared_Files$ mysql -u root -p -f < users_backup_2.sql
Enter password:
ERROR 1396 (HY000) at line 24: Operation CREATE USER failed for 'hr'@'%'
ERROR 1396 (HY000) at line 35: Operation CREATE USER failed for 'inventory'@'%'
ERROR 1396 (HY000) at line 42: Operation CREATE USER failed for 'irene'@'%'
ERROR 1396 (HY000) at line 45: Operation CREATE USER failed for 'management'@'%'
ERROR 1396 (HY000) at line 48: Operation CREATE USER failed for 'mark'@'%'
ERROR 1396 (HY000) at line 51: Operation CREATE USER failed for 'mary'@'%'
ERROR 1396 (HY000) at line 59: Operation CREATE USER failed for 'sales'@'%'
ERROR 1396 (HY000) at line 66: Operation CREATE USER failed for 'sally'@'%'
ERROR 1396 (HY000) at line 69: Operation CREATE USER failed for 'sam'@'%'
ERROR 1396 (HY000) at line 89: Operation CREATE USER failed for 'mysql.infoschema'@'localhost'
ERROR 1396 (HY000) at line 91: Operation CREATE USER failed for 'mysql.session'@'localhost'
ERROR 1396 (HY000) at line 95: Operation CREATE USER failed for 'mysql.sys'@'localhost'
ERROR 1396 (HY000) at line 99: Operation CREATE USER failed for 'root'@'localhost'
```

If you are interested in knowing more about why we have to run this command more than once and why we use the `-f force` command option, come and ask me – it's part of a longer story.

Note again: You **don't need** do to the restore.